

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

Product Description: **Aluminum oxide, 20% in H<sub>2</sub>O, colloidal dispersion**  
Cat No. : **S36593**  
Molecular Formula **Al<sub>2</sub>O<sub>3</sub>**

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
Uses advised against No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address [begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

**CLP Classification - Regulation (EC) No 1272/2008**

**Physical hazards**

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Based on available data, the classification criteria are not met

## Health hazards

Based on available data, the classification criteria are not met

## Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements

None required

EUH210 - Safety data sheet available on request

## 2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component                                        | CAS No    | EC No     | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                        |
|--------------------------------------------------|-----------|-----------|----------|--------------------------------------------------------------------------------------------------------------------------|
| Water                                            | 7732-18-5 | 231-791-2 | 79.50    | -                                                                                                                        |
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) | 1344-28-1 | 215-691-6 | 20.00    | -                                                                                                                        |
| Nitric acid                                      | 7697-37-2 | 231-714-2 | 0.50     | Ox. Liq. 3 (H272)<br>Met. Corr. 1 (H290)<br>Acute Tox. 3 (H331)<br>Skin Corr. 1A (H314)<br>Eye Dam. 1 (H318)<br>(EUH071) |

| Component   | Specific concentration limits (SCL's)                                                                                                                                                                                                                                             | M-Factor | Component notes |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------|
| Nitric acid | Ox. Liq. 2 :: C>=99%<br>Ox. Liq. 3 :: 65%<=C<99%<br>Acute Tox. 1 (inhal) :: C>=70%<br>Acute Tox. 3 (inhal) ::<br>70%>C>=26.5%<br>Acute Tox. 4 (inhal) ::<br>26.5%>C>=13.25%<br>Skin Corr. 1A :: C>=20%<br>Skin Corr. 1B :: 5%<=C<20%<br>Met. Corr. 1 :: C>=2%<br>EUH071 :: C>=20% | -        | -               |

| Component   | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|-------------|-----------------------|-------------------------|-----------------------------|
| Nitric acid | -                     | -                       | ATE = 2.65 mg/L (vapours)   |

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Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                   |
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur.                                               |
| <b>Self-Protection of the First Aider</b> | No special precautions required.                                                                                        |

### 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                        |
|---------------------------|------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. |
|---------------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

**Suitable Extinguishing Media**  
Not combustible.

**Extinguishing media which must not be used for safety reasons**  
No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

**Hazardous Combustion Products**  
Nitrogen oxides (NO<sub>x</sub>), Fumes of aluminum or aluminum oxide.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required.

### 6.2. Environmental precautions

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Should not be released into the environment. See Section 12 for additional Ecological Information.

## 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal.

## 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 12

**Switzerland - Storage of hazardous substances**

Storage class - SC 10/12  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component                                        | European Union                                             | The United Kingdom                                                                                                                        | France                                                      | Belgium                                                  | Spain                                                                                 |
|--------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------|
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) |                                                            | STEL: 30 mg/m <sup>3</sup> 15 min<br>STEL: 12 mg/m <sup>3</sup> 15 min<br>TWA: 10 mg/m <sup>3</sup> 8 hr<br>TWA: 4 mg/m <sup>3</sup> 8 hr | TWA / VME: 10 mg/m <sup>3</sup><br>(8 heures).              | TWA: 1 mg/m <sup>3</sup> 8 uren                          | TWA / VLA-ED: 10 mg/m <sup>3</sup> (8 horas)<br>VLA-ED: 1 mg/m <sup>3</sup> (8 horas) |
| Nitric acid                                      | STEL: 1 ppm (15min)<br>STEL: 2.6 mg/m <sup>3</sup> (15min) | STEL: 1 ppm 15 min<br>STEL: 2.6 mg/m <sup>3</sup> 15 min                                                                                  | STEL / VLCT: 1 ppm.<br>indicative limit<br>STEL / VLCT: 2.6 | STEL: 1 ppm 15 minuten<br>STEL: 2.6 mg/m <sup>3</sup> 15 | STEL / VLA-EC: 1 ppm (15 minutos).<br>STEL / VLA-EC: 2.6                              |

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|                                                  |                                                                                                          |                                                                                                                                                                                                                                | mg/m <sup>3</sup> . indicative limit                                                                                         | minuten                                                                              | mg/m <sup>3</sup> (15 minutos).                                                                                                                                                   |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Component                                        | Italy                                                                                                    | Germany                                                                                                                                                                                                                        | Portugal                                                                                                                     | The Netherlands                                                                      | Finland                                                                                                                                                                           |
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) |                                                                                                          | TWA: 1.25 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 10 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 4 mg/m <sup>3</sup> (8 Stunden). MAK<br>TWA: 1.5 mg/m <sup>3</sup> (8 Stunden). MAK | TWA: 1 mg/m <sup>3</sup> 8 horas                                                                                             |                                                                                      |                                                                                                                                                                                   |
| Nitric acid                                      | STEL: 1 ppm 15 minuti. Short-term<br>STEL: 2.6 mg/m <sup>3</sup> 15 minuti. Short-term                   | TWA: 1 ppm (8 Stunden). AGW -<br>TWA: 2.6 mg/m <sup>3</sup> (8 Stunden). AGW -                                                                                                                                                 | STEL: 1 ppm 15 minutos<br>STEL: 2.6 mg/m <sup>3</sup> 15 minutos<br>TWA: 2 ppm 8 horas                                       | STEL: 1.3 mg/m <sup>3</sup> 15 minuten                                               | TWA: 0.5 ppm 8 tunteina<br>TWA: 1.3 mg/m <sup>3</sup> 8 tunteina<br>STEL: 1 ppm 15 minuutteina<br>STEL: 2.6 mg/m <sup>3</sup> 15 minuutteina                                      |
| Component                                        | Austria                                                                                                  | Denmark                                                                                                                                                                                                                        | Switzerland                                                                                                                  | Poland                                                                               | Norway                                                                                                                                                                            |
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) | MAK-KZGW: 10 mg/m <sup>3</sup> 15 Minuten<br>MAK-TMW: 5 mg/m <sup>3</sup> 8 Stunden                      | TWA: 5 mg/m <sup>3</sup> 8 timer<br>TWA: 2 mg/m <sup>3</sup> 8 timer<br>STEL: 10 mg/m <sup>3</sup> 15 minutter<br>STEL: 4 mg/m <sup>3</sup> 15 minutter                                                                        | STEL: 24 mg/m <sup>3</sup> 15 Minuten<br>TWA: 3 mg/m <sup>3</sup> 8 Stunden<br>TWA: 10 mg/m <sup>3</sup> 8 Stunden           | TWA: 2.5 mg/m <sup>3</sup> 8 godzinach<br>TWA: 1.2 mg/m <sup>3</sup> 8 godzinach     | TWA: 10 mg/m <sup>3</sup> 8 timer<br>STEL: 20 mg/m <sup>3</sup> 15 minutter. set equal to the limit value for Nuisance dust;value calculated                                      |
| Nitric acid                                      | MAK-KZGW: 1 ppm 15 Minuten<br>MAK-KZGW: 2.6 mg/m <sup>3</sup> 15 Minuten                                 | STEL: 1 ppm 15 minutter<br>STEL: 2.6 mg/m <sup>3</sup> 15 minutter                                                                                                                                                             | STEL: 2 ppm 15 Minuten<br>STEL: 5 mg/m <sup>3</sup> 15 Minuten<br>TWA: 2 ppm 8 Stunden<br>TWA: 5 mg/m <sup>3</sup> 8 Stunden | STEL: 2.6 mg/m <sup>3</sup> 15 minutach<br>TWA: 1.4 mg/m <sup>3</sup> 8 godzinach    | TWA: 2 ppm 8 timer<br>TWA: 5 mg/m <sup>3</sup> 8 timer<br>STEL: 4 ppm 15 minutter. value calculated<br>STEL: 10 mg/m <sup>3</sup> 15 minutter. value calculated                   |
| Component                                        | Bulgaria                                                                                                 | Croatia                                                                                                                                                                                                                        | Ireland                                                                                                                      | Cyprus                                                                               | Czech Republic                                                                                                                                                                    |
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) |                                                                                                          | TWA-GVI: 10 mg/m <sup>3</sup> 8 satima. total dust, inhalable particles<br>TWA-GVI: 4 mg/m <sup>3</sup> 8 satima. respirable dust                                                                                              |                                                                                                                              |                                                                                      |                                                                                                                                                                                   |
| Nitric acid                                      | STEL : 1 ppm<br>STEL : 2.6 mg/m <sup>3</sup>                                                             | STEL-KGVI: 1 ppm 15 minutama.<br>STEL-KGVI: 2.6 mg/m <sup>3</sup> 15 minutama.                                                                                                                                                 | STEL: 1 ppm 15 min<br>STEL: 2.6 mg/m <sup>3</sup> 15 min                                                                     | STEL: 1 ppm<br>STEL: 2.6 mg/m <sup>3</sup>                                           | TWA: 1 mg/m <sup>3</sup> 8 hodinách.<br>Ceiling: 2.5 mg/m <sup>3</sup>                                                                                                            |
| Component                                        | Estonia                                                                                                  | Gibraltar                                                                                                                                                                                                                      | Greece                                                                                                                       | Hungary                                                                              | Iceland                                                                                                                                                                           |
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) | TWA: 10 mg/m <sup>3</sup> 8 tundides. total dust<br>TWA: 4 mg/m <sup>3</sup> 8 tundides. respirable dust |                                                                                                                                                                                                                                | TWA: 10 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup>                                                                        | TWA: 5 mg/m <sup>3</sup> 8 órában. AK Al<br>TWA: 2 mg/m <sup>3</sup> 8 órában. AK Al | TWA: 10 mg/m <sup>3</sup> 8 klukkustundum. Al<br>Ceiling: 20 mg/m <sup>3</sup> Al                                                                                                 |
| Nitric acid                                      | STEL: 1 ppm 15 minutites.<br>STEL: 2.6 mg/m <sup>3</sup> 15 minutites.                                   | STEL: 1 ppm 15 min<br>STEL: 2.6 mg/m <sup>3</sup> 15 min                                                                                                                                                                       | STEL: 1 ppm<br>STEL: 2.6 mg/m <sup>3</sup>                                                                                   | STEL: 2.6 mg/m <sup>3</sup> 15 percekbén. CK                                         | STEL: 1 ppm<br>STEL: 2.6 mg/m <sup>3</sup>                                                                                                                                        |
| Component                                        | Latvia                                                                                                   | Lithuania                                                                                                                                                                                                                      | Luxembourg                                                                                                                   | Malta                                                                                | Romania                                                                                                                                                                           |
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) | TWA: 6 mg/m <sup>3</sup>                                                                                 | TWA: 5 mg/m <sup>3</sup> inhalable fraction IPRD Al<br>TWA: 2 mg/m <sup>3</sup> respirable fraction IPRD Al                                                                                                                    |                                                                                                                              |                                                                                      | TWA: 2 mg/m <sup>3</sup> 8 ore<br>TWA: 3 mg/m <sup>3</sup> 8 ore<br>TWA: 1 mg/m <sup>3</sup> 8 ore<br>STEL: 5 mg/m <sup>3</sup> 15 minute<br>STEL: 10 mg/m <sup>3</sup> 15 minute |

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|             |                                                                                         |                                            |                                                                  |                                                                |                                                                |
|-------------|-----------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|
|             |                                                                                         |                                            |                                                                  |                                                                | STEL: 3 mg/m <sup>3</sup> 15 minute                            |
| Nitric acid | STEL: 1 ppm<br>STEL: 2.6 mg/m <sup>3</sup><br>TWA: 0.78 ppm<br>TWA: 2 mg/m <sup>3</sup> | STEL: 1 ppm<br>STEL: 2.6 mg/m <sup>3</sup> | STEL: 1 ppm 15 Minuten<br>STEL: 2.6 mg/m <sup>3</sup> 15 Minuten | STEL: 1 ppm 15 minuti<br>STEL: 2.6 mg/m <sup>3</sup> 15 minuti | STEL: 1 ppm 15 minute<br>STEL: 2.6 mg/m <sup>3</sup> 15 minute |

| Component                                        | Russia                                                                                                                                                                                                 | Slovak Republic                                                                       | Slovenia                                                                                                                   | Sweden                                                                                                                                                        | Turkey                                                         |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) | TWA: 6 mg/m <sup>3</sup> 0043 in the form of disintegration aerosol<br>TWA: 1 mg/m <sup>3</sup> 0045 containing up to 20% Cr <sub>2</sub> O <sub>3</sub> ;catalyst IM-2201<br>MAC: 3 mg/m <sup>3</sup> | TWA: 4 mg/m <sup>3</sup> inhalable dust<br>TWA: 1.5 mg/m <sup>3</sup> respirable dust |                                                                                                                            | TLV: 5 mg/m <sup>3</sup> 8 timmar.<br>AI NGV<br>TLV: 2 mg/m <sup>3</sup> 8 timmar.<br>AI NGV                                                                  |                                                                |
| Nitric acid                                      | Skin notation<br>MAC: 2 mg/m <sup>3</sup>                                                                                                                                                              | Ceiling: 2.6 mg/m <sup>3</sup>                                                        | TWA: 1 ppm 8 urah<br>TWA: 2.6 mg/m <sup>3</sup> 8 urah<br>STEL: 1 ppm 15 minutah<br>STEL: 2.6 mg/m <sup>3</sup> 15 minutah | Binding STEL: 1 ppm 15 minuter<br>Binding STEL: 2.6 mg/m <sup>3</sup> 15 minuter<br>TLV: 0.5 ppm 8 timmar.<br>NGV<br>TLV: 1.3 mg/m <sup>3</sup> 8 timmar. NGV | STEL: 1 ppm 15 dakika<br>STEL: 2.6 mg/m <sup>3</sup> 15 dakika |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                                                               | Fresh water       | Fresh water sediment | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture) |
|-------------------------------------------------------------------------|-------------------|----------------------|--------------------|------------------------------------|--------------------|
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> )<br>1344-28-1 ( 20.00 ) | PNEC = 0.3136µg/L |                      | PNEC = 3.136µg/L   | PNEC = 20mg/L                      |                    |

## 8.2. Exposure controls

### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

Protective gloves

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| Glove material                                      | Breakthrough time                    | Glove thickness | EU standard | Glove comments        |
|-----------------------------------------------------|--------------------------------------|-----------------|-------------|-----------------------|
| Natural rubber<br>Nitrile rubber<br>Neoprene<br>PVC | See manufacturers<br>recommendations | -               | EN 374      | (minimum requirement) |

**Skin and body protection** Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particle filter

**Small scale/Laboratory use** Maintain adequate ventilation

**Environmental exposure controls** No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid dispersion        |                                          |
| <b>Appearance</b>                              |                          |                                          |
| <b>Odor</b>                                    | Odorless                 |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flammability (liquid)</b>                   | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | No information available | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>pH</b>                                      | No information available |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Water Solubility</b>                        | Miscible                 |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Nitric acid                                    | -2.3                     |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | No data available        |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)  |                                          |

### 9.2. Other information

**Molecular Formula** Al<sub>2</sub> O<sub>3</sub>

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## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

#### Hazardous Polymerization

No information available.

#### Hazardous Reactions

None under normal processing.

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

### 10.5. Incompatible materials

None known.

### 10.6. Hazardous decomposition products

Nitrogen oxides (NO<sub>x</sub>). Fumes of aluminum or aluminum oxide.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

No data available

Inhalation

No data available

#### Toxicology data for the components

| Component                                        | LD50 Oral                                     | LD50 Dermal | LC50 Inhalation                        |
|--------------------------------------------------|-----------------------------------------------|-------------|----------------------------------------|
| Water                                            | -                                             | -           | -                                      |
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) | > 5000 mg/kg. ( Rat )<br>(OECD Guideline 401) | -           | > 2.3 mg/l 4 h<br>(OECD Guideline 403) |
| Nitric acid                                      | -                                             | -           | LC50 = 2500 ppm. (Rat) 1h              |

| Component   | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|-------------|-----------------------|-------------------------|-----------------------------|
| Nitric acid | -                     | -                       | ATE = 2.65 mg/L (vapours)   |

#### (b) skin corrosion/irritation;

No data available

#### (c) serious eye damage/irritation;

No data available

#### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

#### (e) germ cell mutagenicity;

No data available



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(f) carcinogenicity; No data available

The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component                                        | EU | UK | Germany             | IARC |
|--------------------------------------------------|----|----|---------------------|------|
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) |    |    | Cat. 2 (Fibre dust) |      |

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects .

### 12.2. Persistence and degradability

**Persistence** Miscible with water, Persistence is unlikely, based on information available.

**12.3. Bioaccumulative potential** Bioaccumulation is unlikely

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| Nitric acid | -2.3    | No data available             |

### 12.4. Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

**12.5. Results of PBT and vPvB assessment** No data available for assessment.

### 12.6. Endocrine disrupting properties

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

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## 12.7. Other adverse effects

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Ozone Depletion Potential**

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products**

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

**Contaminated Packaging**

Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

**European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used.

**Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

Not regulated

14.1. UN number

14.2. UN proper shipping name

14.3. Transport hazard class(es)

14.4. Packing group

ADR

Not regulated

14.1. UN number

14.2. UN proper shipping name

14.3. Transport hazard class(es)

14.4. Packing group

IATA

Not regulated

14.1. UN number

14.2. UN proper shipping name

14.3. Transport hazard class(es)

14.4. Packing group

14.5. Environmental hazards

No hazards identified

14.6. Special precautions for user

No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

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## 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                                        | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|--------------------------------------------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Water                                            | 7732-18-5 | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) | 1344-28-1 | 215-691-6 | -      | -   | X     | X    | KE-01012 | X    | X    |
| Nitric acid                                      | 7697-37-2 | 231-714-2 | -      | -   | X     | X    | KE-25911 | X    | X    |

| Component                                        | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|--------------------------------------------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Water                                            | 7732-18-5 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) | 1344-28-1 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Nitric acid                                      | 7697-37-2 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component                                        | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------------------------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Water                                            | 7732-18-5 | -                                                                   | -                                                                             | -                                                                                                     |
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) | 1344-28-1 | -                                                                   | -                                                                             | -                                                                                                     |
| Nitric acid                                      | 7697-37-2 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Seveso III Directive (2012/18/EC)

| Component                                        | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Water                                            | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           |
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) | 1344-28-1 | Not applicable                                                                            | Not applicable                                                                           |
| Nitric acid                                      | 7697-37-2 | Not applicable                                                                            | Not applicable                                                                           |

### Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

### Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

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## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

**WGK Classification** Water endangering class = non-hazardous to waters (self classification)

| Component                                        | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|--------------------------------------------------|---------------------------------------|-------------------------|
| Aluminum oxide (Al <sub>2</sub> O <sub>3</sub> ) | nwg                                   |                         |
| Nitric acid                                      | WGK1                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                         | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Nitric acid<br>7697-37-2 ( 0.50 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H272 - May intensify fire; oxidizer

H290 - May be corrosive to metals

H314 - Causes severe skin burns and eye damage

H318 - Causes serious eye damage

EUH071 - Corrosive to the respiratory tract

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

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## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Prepared By** Health, Safety and Environmental Department

**Revision Date** 05-Mar-2024

**Revision Summary** New emergency telephone response service provider.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**